Smart meter installations in Maharashtra

cropped up, including erroneous meter readings and billing, no proper billing cycle, no safety measures like anti-theft infrastructure or leakage detection, and high operational expenses because of non-optimal battery life and frequent maintenance.

The response to all these challenges came with the installation of smart AMI ultrasonic water meters powered by LoRaWAN technology, replacing older meter readers. The project was executed by SenRa. Hosseini says, "Solutions such as LPWANs based on LoRaWAN protocol allow cities to reduce operational costs dramatically while still obtaining the necessary data to make better decisions."

Dhananjay Sharma, chief operational officer, Senra, explains, "The city selected 41 sites within its limits where they requested to install the new AMI meters. The infrastructure for LoRaWAN network was set up in two locations, selected based on accessibility of site, power availability, backhaul capabilities, elevation above sea-level and ground-level, obstructions and clear line of site. The aim of the meter was to provide end-to-end automated mechanisms including wireless communication, secure data transfer and real-time analytics."

On successful installation, city administratives, utilities company and water distribution company reaped major benefits. First, the smartmeters provide real-time visualisation of data related to water distribution and consumption.

Second, administratives become aware of higher consumption regions and of the fact that water consumption triples during holidays, allowing them to plan their supply accordingly and reduce water wastage.

Daily and monthly water usage analytics enable them to generate accurate bills. Unusual behaviour in water lines warns them of potential leakage or service required, enabling predictive maintenance.

Additionally, there has been a massive reduction in costs pertaining to operations, including substantially reduced investment in manual resources, no frequent maintenance of infrastructure and battery longevity. A long battery life guarantees a vigilance-free functioning up to 12 years.

The infrastructure has been made theft-proof with central alarms. Overall, water distribution has become consistent, with greatly improved satisfaction from the citizens.

Making LPWAN mainstream in India

LPWAN technology has not yet reached its full potential in India. Industry experts share their concerns over India's lacking hardware production ecosystem.

Nambiar says, "Currently, fewer than 50 towns have LPWAN coverage. Even though LPWANs utilise the unlicensed spectrum, network coverage and deployment of LPWAN infrastructure like gateways and base stations depend on public and private investments.

"LPWAN adoption in the enterprise segment is expected to grow fast. However, another challenge is that a significant share of LPWAN implementations will depend on the pace and scale at which the government and public private partnership projects are awarded."

LoRa alliance, Sigfox and other administrative bodies in India are working towards addressing these pain points. However, much ground remains to be covered. EFY

—Paromik Chakraborty, technical journalist, EFY